

Lecture (18): Visual Pathway

■ Structure of the retina:

It consists of **10 layers** (from outwards, inwards):

- 1- **Pigmented epithelium** is the most superficial (**outermost**) layer of retina. Its function is maintaining retinal-blood barrier, has a phagocytic property and absorbs excess light.
- 2- **Photoreceptors (Rods & cones)** lying **closest** to the pigmented epithelium.
 - **Rods:**
 - Distributed throughout the **periphery** of the retina.
 - Effective in dim light (scotopic vision).
 - Contain photochemical (**rhodopsin**).
 - In each retina, the number of rods is about **100 million**
 - **Cons:**
 - Concentrated in the **fovea centralis** of the retina.
 - Responsible for **high acuity color (photopic) vision**.
 - Effective in daylight.
 - There are **3** different **opsins** in the cone cells (**cone pigments; red cones:** long wavelength, **green cones:** medium-wavelength and **blue cones:** short wavelength).
 - In each retina, the number of cones is about **6-7 million**
- 3- **Outer limiting membrane.**
- 4- **Rod & cone cells** (outer nuclear layer).
- 5- **Outer plexiform layer** (synaptic connections between rod, cone cells and bipolar cells).
- 6- **Bipolar cells, 1st order neuron (inner nuclear layer).**

There are two populations of rod and cone bipolar cells, called 'on' and 'off' bipolar cells.
- 7- **Inner plexiform layer** (synaptic connections between bipolar cells and ganglionic cells).
- 8- **Ganglion cell layer (2nd order neuron).** There are numerous types of ganglion cells, their axons pass tangentially to come together at the optic disc to form the optic nerve.
- 9- **Axonal layer of ganglion cells (nerve fibers),** they form the optic nerve.
- 10- **Inner limiting membrane** separates the retina from the vitreous body.
 - **A detached retina** is said to occur when the pigment epithelial cell layer comes away from the photoreceptor layer.
 - **Optic disc:** The circular area in the back of the inside of the eyeball where the optic nerve connects to the retina. No photoreceptors at optic disc (**blind**

spot).

- **Fovea centralis:**

- is a specialized area (thinnest) of the retina.
- It is a depression located within the center of **macula lutea**.
- It has high density of **cones only, no rods**.
- It is responsible for our central (highest visual acuity) vision and colour vision.
 - **Macula lutea** is a rounded yellowish area of the retina situated lateral to optic disc.

Blood supply of retina:

- Outer layers (including rods & cones) are supplied from **short posterior ciliary arteries** from ophthalmic artery.
- Inner layers (Bipolar & ganglion cells) are supplied by **central retinal artery**.

- ▣ **Optic nerve:**

- Each optic nerve containing 0.8–1.5 million axons of retinal ganglion cells. - **It** travels through optic canal to form the optic chiasma.

- ▣ **Optic chiasma:** (point of decussation):

- is an **X-shaped structure** at the base of the brain anterior to the pituitary gland.
- The axons of ganglion cells of each **nasal hemiretina decussate** in the optic chiasma and enter the contralateral optic tract, whereas those of each **temporal hemiretina** remain in the same side and enter the ipsilateral optic tract.

- ▣ **Optic tract:**

- Contains axons from the **ipsilateral temporal hemiretina** and the **contralateral nasal hemiretina**.
- Most of the axons in the optic tract travel via the lateral root of the optic tract to terminate in the **lateral geniculate nucleus (LGN)** of the thalamus. However, **about 10% of the axons** from the medial root terminate in:
 - 1- **Superior colliculus of the midbrain** controlling the movement of extraocular muscles.
 - 2- **Pretectal nucleus of the midbrain** controlling the pupillary light reflexes.
 - 3- **Suprachiasmatic nucleus of the hypothalamus** where they are involved in the setting of circadian rhythms.

- ▣ **LGN (3rd order neuron):**

The **LGN** has **six** major layers, which receive inputs from retinal ganglion cells (**layers 1, 4 and 6 from the contralateral eye and layers 2, 3 and 5 from the ipsilateral eye**) which project to the visual cortex.

■ Optic radiation (geniculocalcarine tract):

- Are axons of **LGN** projecting to the **primary visual cortex**.
 - The **fibres in the optic radiation** take two different routes to the visual cortex:
- 1- **Upper optic radiation** carries fibres from the superior retinal quadrants. It travels through the parietal lobe to reach the visual cortex.
 - 2- **Lower optic radiation** carries fibres from the inferior retinal quadrants. It travels through the **temporal lobe**, via a pathway known as **Meyer's loop**, to reach the visual cortex.

Primary visual cortex, striate cortex (V1, Area 17):

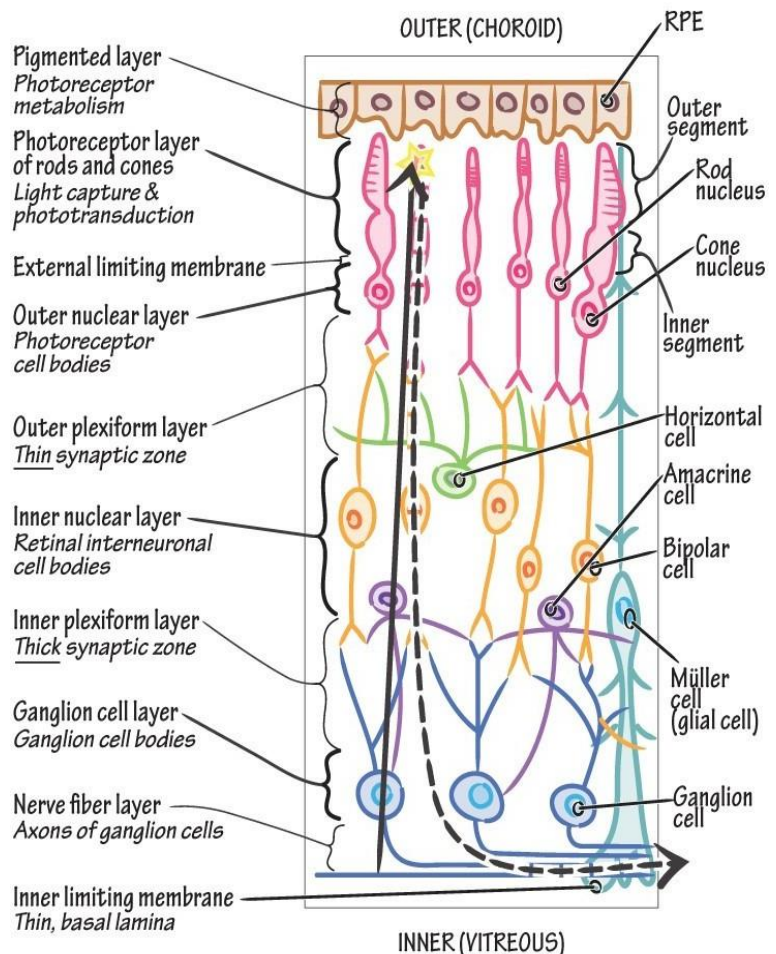
- It is in the **calcarine sulcus** on the medial surface of the **occipital lobe** extending to **occipital pole**.

- It is formed by **six layers of neurons**
- Receives input from layers of the **LGB**.

- It is important for binocular vision, perception of movement and perception of color & shape. Extrastriate Visual Cortex (V2– V5):

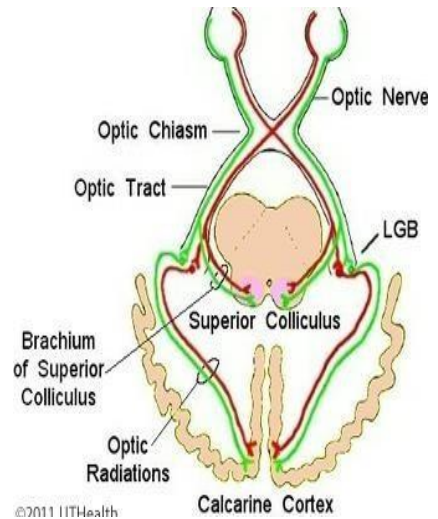
- The parts of the extrastriate visual cortex are composed of Brodmann areas **18 and 19**.
- They receive inputs from the primary visual area via:
 - 1- The **ventral stream** passes downward from the occipital lobe to the temporal lobe and carries information about object identification including shape, contrast, and colour.
 - 2- The **dorsal stream** passes upward from the occipital lobe to the parietal lobe and carries information about an object's location and motion (visually guided movements).

- **Lesion to areas 18 & 19** causes **visual agnosia**, inability to recognize visual objects.

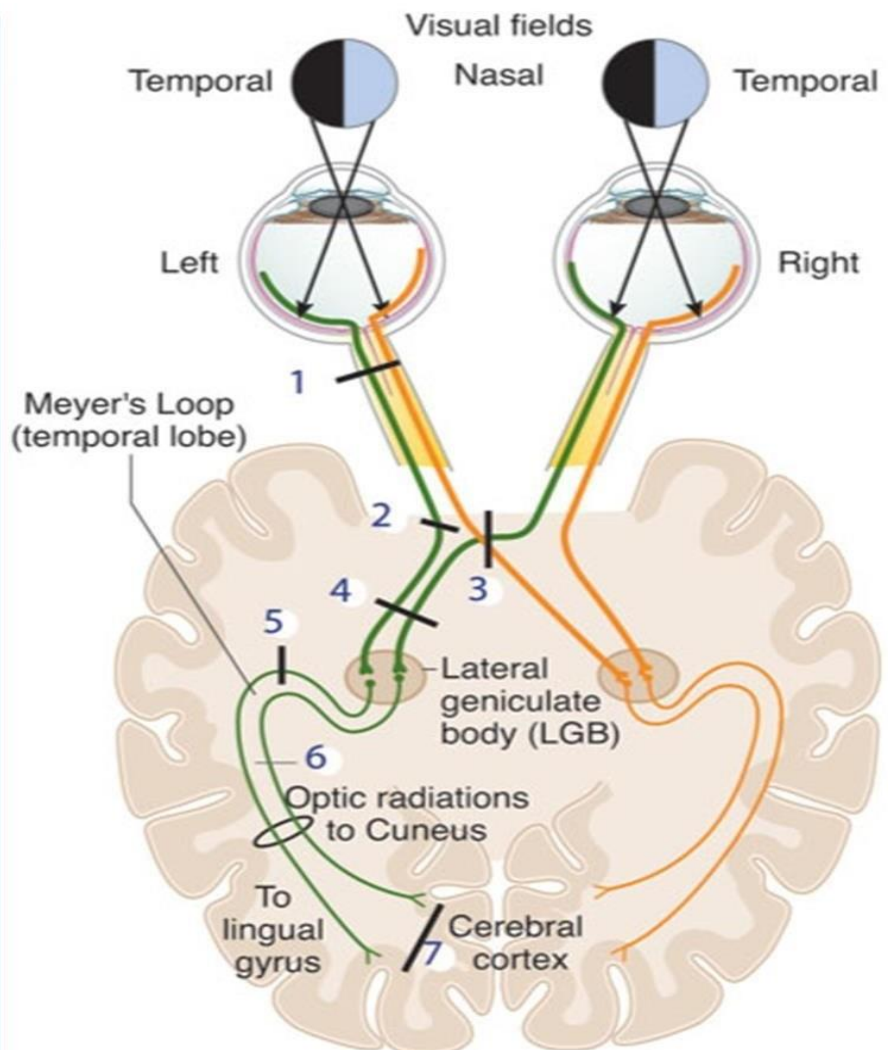


■ Visual field defects:

- A. **Anopia (loss of vision) in left eye:** complete section left optic nerve.
- B. **Lt nasal hemianopia:** unilateral lesion to temporal fibers of optic chiasma
- C. **Bitemporal hemianopia:** by compression of the chiasma by an adenoma, a benign tumour of the pituitary.
- D. **Right homonymous hemianopia:** section of the left optic tract.
- E. **Right homonymous upper quadrantanopia:** damage to the **right Meyer's loop**.
- F. **Right homonymous hemianopia:** damage to the optic radiation.
- G. **Right homonymous lower quadrantanopia:** parietal lobe lesion.
- H. **Homonymous hemianopia with macular sparing:** lesion above or below the calcarine sulcus, due to a stroke in the posterior cerebral artery.



Defects	
1	Anopia of left eye
2	Left nasal hemianopia
3	Bitemporal heteronymous hemianopia
4	Right homonymous hemianopia
5	Right homonymous superior quadrantanopia
6	Right homonymous inferior quadrantanopia
7	Right homonymous hemianopia with macular sparing



Pupillary Light Reflex Pathway:

- The light stimulates the retinal ganglionic cells. The impulses travel through the optic nerve (CN II), which projects bilaterally to the **pretectal nucleus** in the midbrain and then projects to the **Edinger-Westphal nucleus** bilaterally.
- The efferent limb is directed by the **oculomotor nerve (CN III)**. The Edinger-Westphal nucleus (preganglionic parasympathetic) relays to the **ciliary ganglion**.
- Miosis on the same side (direct light reflex) & opposite side (Consensual light reflex) to protect itself when light intensity increases.

Accommodation (near) reflex:

- The information is sent back through **efferent fibers** to the **pretectal nucleus** and then to the **Edinger-Westphal nucleus**, which contains preganglionic parasympathetic neurons whose axons travel in the oculomotor nerve to the orbit where they form a synapse in the **ciliary ganglion**.
- **Contraction of the sphincter pupillae and the ciliary muscle**
- **Convergent eye movements** caused by **contraction of the medial recti muscles** which are innervated by the **oculomotor nerve**.

Lecture (19): Auditory and vestibular pathways

Auditory pathway:

- ▣ The **ear pinna** and the **external auditory meatus** collect sound waves into a unidirectional wave towards the **tympanic membrane**.
- ▣ Sound waves cause vibrations of the **tympanic membrane**. The louder the sound the bigger the vibration, the lower pitch the sound the slower the vibration.
 - ▣ **Amplification** of sound waves occur in middle ear by **3 bony ossicles (malleus, incus & stapes)** which connect the **tympanic membrane** to the **oval window** towards the inner ear.
 - ▣ The **handle of the malleus** attaches with the tympanic membrane, and **the malleus** articulates with **the Incus**.
 - ▣ The **incus** articulates with the stapes, which impacts onto the **oval window**. The stapes moves like a piston, and causes the oval window to move in and out with sound waves.
 - ▣ In the inner ear, the sound waves are sent up the **scala vestibuli** to the apex of the cochlear duct (the helicotrema). Here it continues back down in the **scala tympani**.
 - ▣ The scala vestibuli and scala media are separated by **Reissner's membrane**.
 - ▣ Scala media and scala tympani below are separated by the **basilar membrane**.